

RAJEEV GOEL

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Seeking a Summer 2025 research internship in 3D Generative AI to advance methods and contribute to innovative projects.

EDUCATION

Arizona State University Doctor of Philosophy, Computer Science.	<i>Aug 2023 - Present</i> GPA: 3.76
Arizona State University Master of Science, Computer Science	<i>May 2023</i> GPA: 3.75
Thapar Institute of Engineering and Technology Bachelor of Engineering, Electronics and Communications Engineering.	<i>May 2020</i> CGPA: 7.84

RESEARCH PUBLICATIONS

Deep Geometric Moments Promote Shape Consistency in Text-to-3D Generation Feb 2025
*Utkarsh Nath, **Rajeev Goel**, Eun Som Jeon, Changhoon Kim, Kyle Min, Yezhou Yang, Yingzhen Yang and Pavan Turaga.*
WACV 2025 [[Paper](#)][[Project Page](#)]

- MT3D: A 3D Gaussian-based 2D lifting technique that leverages a high-fidelity 3D object to explicitly infuse geometric knowledge into text-to-3D image generation.
- Utilizes ControlNet and Geometric moment analysis to optimize and refine the shape and structure of 3D objects, effectively alleviating the Janus problem.
- Surpassing other state-of-the-art text-to-3D generators, MT3D significantly reduces geometric inconsistencies, delivering superior shape, high-fidelity, and enhanced photorealism.

Learning Low-Rank features for Thorax Disease Classification Dec 2024
Yancheng Wang, **Rajeev Goel***, Utkarsh Nath, AC Silva, Teresa Wu and Yingzhen Yang.*
NeurIPS 2024 [[Paper](#)][[Code](#)] *Collaboration with Mayo Clinic*

- Implemented a separable approximation of the Truncated Nuclear Norm, enabling the optimization of the LRFL training loss using standard SGD, and enhancing the generalization ability of the model.
- Achieves SOTA results for three standard medical benchmark datasets such as NIH-ChestX-ray14, CheXpert and CovidX using low-rank regularization.
- Low-rank model achieves 1-2% higher AUC than its counterparts under a low-data regime.
- Enhanced low-rank model performance by 0.6% through Generative Data Augmentation.

IDNet: A Novel ID Dataset via Few-Shot and Quality-Driven Synthetic Data Generation Dec 2024
*Lulu Xie, Yancheng Wang, Hong Guan, Soham Nag, **Rajeev Goel**, Niranjan Swamy, Yingzhen Yang, Chaowei Xiao, Jonathan Prisby, Ross Maciejewski, and Jia Zou.*
BigData 2024 [[Paper](#)] *Collaboration with the United States Department of Homeland Security*

- Developed a pipeline for realistic mobile document generation using parameterized ID replacement.
- Leveraged advanced models like Grounding DINO and SAM for precise object detection and segmentation, and employed Deep Image Blending with SSIM loss to enhance blending realism, producing realistic, artifact-free outputs for the IDNet dataset.

WORK EXPERIENCE

IBM India Pvt. Ltd Bengaluru, India
Blockchain Application Developer Nov 2020 - Nov 2021

- Reduced costs by 15% by migrating client databases from Cloudant to CouchDB.
- Researched Hyperledger Fabric migration, and automated backend server management, improving efficiency by 20%.

IBM India Pvt. Ltd Bengaluru, India
Full Stack Development Intern Jan 2020 - June 2020

- Developed REST APIs with Java Spring Boot to integrate systems, and designed responsive UI components using HTML, CSS, and ReactJS, optimizing webpage load times by 15%.

TECHNICAL SKILLS

Programming Languages: Python, Bash
Libraries and Frameworks: PyTorch, NumPy, Pandas, OpenCV, Matplotlib
Tools and Platforms: Linux, Git, AWS, Weights and Biases

TEACHING EXPERIENCE

Teaching Assistant for Knowledge Representation and Reasoning (CSE 579), Foundation of Machine Learning (CSE 475), and Introduction to Computer Science (CS 194) .